

# Errata — *Wine Production Forecasting Using Rule-Based Machine Learning*

**Hugo Filipe Queiróz Nogueira** · MECD, FEUP · June 2026 Corrections identified after submission of the version deposited for defense. Page numbers refer to the submitted PDF (*Tese Hugo Nogueira — final version*). All corrections are incorporated in the definitive version.

| # | Page | Location                                                    | Submitted text                                                                     | Correction                                                                                                                                                                                                                                                       |
|---|------|-------------------------------------------------------------|------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | 36   | Ch.3, §"CarenR: Distribution Rules via the CAREN Framework" | "CarenR is built on the CAREN association rule engine, <b>originally developed</b> | The CAREN engine was developed by <b>Azevedo (2003)</b> ; Jorge, Azevedo & Pereira (2006) introduced the <b>distribution-rule</b> framework on top of it. Corrected to: "...originally developed by Azevedo (2003) and extended to the discovery of distribution |

|   |    |                                   |                                |                                                                                                                                                                                                           |
|---|----|-----------------------------------|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   |    |                                   | by Jorge et al. (2006)"        | rules in numeric data by Jorge et al. (2006)."<br>New reference added:<br>Azevedo, P. J. (2003), <i>CAREN – A Java based Apriori implementation for classification purposes</i> , Tech. Report, U. Minho. |
| 2 | 99 | Bibliography, Jorge et al. (2006) | DOI<br>10.1007/11871637_**25** | Correct DOI is<br>10.1007/11871637_**26**<br>(PKDD 2006, LNCS/LNAI vol. 4213, pp. 247–258, Springer).<br>Volume and pages added.                                                                          |
|   |    |                                   |                                | CAREN performs an <b>Apriori-</b>                                                                                                                                                                         |

|   |            |                                          |                                                                                                                                                            |                                                                                                                                                                                                                                                                                          |
|---|------------|------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3 | 28 / 36–38 | Ch.2 (p.28)<br>and Ch.3<br>Step 3 (p.36) | Search<br>described as<br><b>"beam<br/>search"</b><br>(Ch.2) and<br><b>"breadth-<br/>first ... most<br/>promising<br/>single<br/>conditions"</b><br>(Ch.3) | <b>style<br/>levelwise<br/>search<br/>pruned by<br/>minimum<br/>support</b><br>(exhaustive<br>over the<br>support-<br>frequent<br>region, not a<br>heuristic<br>beam). Both<br>passages<br>harmonised<br>to this<br>description.                                                         |
|   |            |                                          | Stated as<br><b>"15%" / "≥</b><br>15% of<br>observations<br>" in Ch.2,                                                                                     | Actual value<br>used in the<br>pipeline is<br><b>20%</b><br><code>( carenr_m<br/>in_sup &lt;-<br/>0.20 in<br/>3_carenR<br/>__applica<br/>tion.R</code><br>and<br><code>6_scenar<br/>io_valida<br/>tion.R )</code> ,<br>consistent<br>with the<br>results in<br>Ch.4–6. All<br>statements |

|   |                                 |                           |                                                                  |                                                                                                                                                                                                                             |
|---|---------------------------------|---------------------------|------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4 | 28, 36, 38, 49, 100+ (Appx A/B) | Minimum support threshold | Ch.3 (Step 5 and weighting paragraph), Appendix A and Appendix B | corrected to 20%.<br>Derived counts corrected accordingly ("≈18 years out of 89"; weighting example "≈1.4×").<br><i>This propagates a correction already logged in Appendix E to the chapters where it had been missed.</i> |
|   |                                 | Bibliography,             | Title given as<br>***Contrast Association Rules for Extraordinar | Backronym removed (title corrected to a neutral description of the R package).<br>Printed "TODO" notes removed from this and the Fraga                                                                                      |

|   |     |                       |                                                    |                                                                                                                                                                                                                                          |
|---|-----|-----------------------|----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5 | 100 | Moreira et al. (2022) | y Nodes**", plus a printed "TODO: confirm..." note | (2024) and Lopes (2024) entries. <i>Exact bibliographic metadata for Moreira (2022), Fraga (2024) and Lopes (2024) to be confirmed with supervisors.</i>                                                                                 |
|   |     |                       |                                                    | Not supported by the rule output ( rules_in interpreted _rvv.csv ). Across the 20 RVV rules the <b>LAI family appears in 4/20 and LAI at harvest specifically in 2/20</b> ; the dominant RVV features are <b>heat-stress days (up to</b> |

|   |                                                                                                                                            |                           |                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                      |
|---|--------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 6 | <p>Abstract;<br/>Ch.4 (Table 4.9 cross-algorithm, RVV feature-frequency Table, §"RVV top rules", M5Rules comparison); Ch.2; Appendix C</p> | <p>RVV LAI over-claim</p> | <p>LAI described as the <b>dominant RVV feature, "present in all 20 rules" / "20/20 rules"</b> (Abstract, Table 4.9, Appendix C, Ch.2, Ch.4).</p> | <p><b>50% of rules)</b> and <b>harvest soil water (25%)</b> — as the thesis's own frequency table (Table 4.9) already shows (LAI at harvest is rank 10, 10%). All "all 20 / 20/20 / dominant-LAI" statements corrected to 4/20 (family) or 2/20 (harvest); the Ch.2 example of a CarenR-specific feature changed from LAI (which all three algorithms select) to wet-days post-flowering. Extends the Appendix E</p> |
|---|--------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|  |  |  |  |                                                                    |
|--|--|--|--|--------------------------------------------------------------------|
|  |  |  |  | correction<br>(which had<br>logged 4/20<br>but not<br>propagated). |
|--|--|--|--|--------------------------------------------------------------------|

**Note on nature of corrections.** Items 1–5 are editorial/bibliographic and affect no result, table, test, or conclusion. Item 6 is a **factual correction to a feature-frequency description**: the RVV rule *counts* (20), MAEs, statistical tests, and conclusions are unchanged, but the claim that LAI is the dominant RVV feature was wrong — LAI appears in 4/20 rules, and heat-stress days and soil water are the dominant RVV features (consistent with the thesis's own Table 4.9 and Appendix E). The support-threshold value used in all computations was 20% throughout; item 4 corrects descriptive text that lagged the code, not the analysis itself.